

The Topography of Human-Computer Interaction

Tor-Salve Dalsgaard
Department of Computer Science
University of Copenhagen
Copenhagen, Denmark
torsalve@di.ku.dk

Katerina Cerna
ISDD
Halmstad University
Halmstad, Sweden
katerina.cerna@hh.se

Abstract

We present the Topography of Human-Computer Interaction, a participatory interactive experience that invites conference participants to collaboratively create a material representation of HCI. Drawing on concepts of boundary objects and maker pedagogy, we propose an alternative to traditional field overviews that emphasises collective making over abstract understanding. Participants shape a shared clay topography by placing small pieces of clay to a shared topography that reflect their research positions, creating an evolving landscape through ongoing interaction and conversation. Building on iterations at the Decennial Aarhus Conference and Halmstad University, we present key insights about participant engagement and community-building. For CHI 2026, we introduce refinements to the process, including shared labelling through strings and stickers and a stronger emphasis on the shape of a topography as a landscape. The resulting topography serves as both a physical artefact and a catalyst for reflection on how the CHI community perceives itself.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Human-Computer Interaction, Community, Topography

ACM Reference Format:

Tor-Salve Dalsgaard and Katerina Cerna. 2026. The Topography of Human-Computer Interaction. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799112>

1 Introduction

Making maps of the spaces humans inhabit has a long history, from early geographic representations to contemporary mappings of social, cultural, and intellectual domains. Maps are not neutral depictions of reality; they are shaped by the perspectives, positions, and assumptions of those who create them, and they gain meaning through the relationships they establish between people, places, and ideas. Within human-computer interaction (HCI), the need to gain an overview of the field has repeatedly been addressed through

academic papers (e.g., [1]), essays (e.g., [7, 8]), and textbooks (e.g., [3]). While such representations are valuable, they are necessarily partial and situated in time. Moreover, they often privilege abstraction over lived experience, making it difficult to engage with the field as something felt, negotiated, and collectively shaped.

In this work, we propose a different approach: a participatory, material topography of HCI. Drawing on topography as a metaphor, we invite the CHI community to collaboratively sculpt a shared material representation of the field. Topography represents the physical shape and surface of a landscape. While cartographic perspectives remind us that any mapping is selective and political, topography allows these insights to be enacted materially, through shared, embodied making. Through an interactive demonstrator, participants contribute to an evolving surface that reflects areas of expertise, common ground, and emerging relations, to create the topography of HCI.

We view the topography as a boundary object. A boundary object enables conversations about the field through its interpretive flexibility. Star and Griesemer argue that “the creation and management of boundary objects is key in developing and maintaining coherence across intersecting social worlds” [6, p. 393], enabling the function of the proposed interactive experience as an object of conversation and community-building. We further draw on maker pedagogy, an educational tradition building on DIY cultures and maker movements, where students actively design, build, and iterate tangible artefacts to construct knowledge through making. In doing so, we explore pedagogical possibilities of participatory making [4, 5] in conference settings that extend beyond the traditional field overviews. Rather than offering an objective map of HCI, the topography functions as a boundary object that supports reflection, conversation, and community-building — situated in the specific context of CHI and shaped by those who take part.

2 Past Iterations of the Topography

In the past, we have hosted our demo experience at two previous occasions. First, we organised an interactive experience during the Sixth Decennial Aarhus Conference [2], in which participants placed colourful plasteline (a type of clay that does not harden) on a round table. Participants were instructed to reflect on their space in HCI, whether it be through contribution, theoretical commitment, or subfield, express this belonging through a small piece of clay, and place their clay in relation to existing clay formations (Figure 1a). The participants also had the opportunity to raise a small flag with a keyword describing their space (Figure 1b). We placed a poster next to the table, providing information about the project, instructions on how to work with the clay, and three scales with which we asked evaluative questions about the experience (Figure 1c). In addition, we documented the topography through photos of the table and

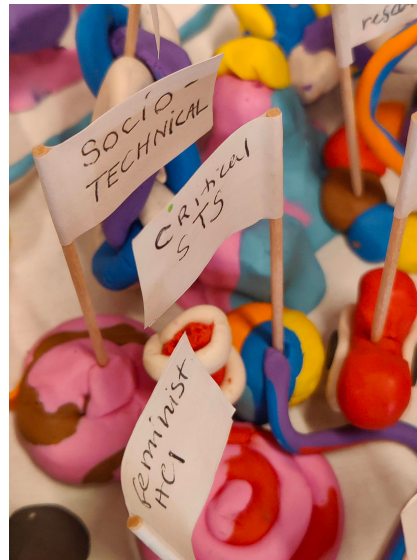


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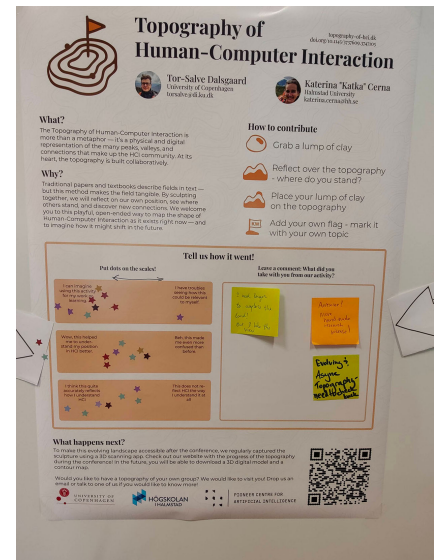
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ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3799112>



(a) Top down view of the table.



(b) Detailed view on clay and flags.



(c) An interactive poster.

Figure 1: The topography at the Sixth Decennial Aarhus Conference.

field notes of impressions and conversations around the table. At the end of the demo session, we made a 3D scan of the topography, available for download at topography-of-hci.dk.

The process of creation of this first topography was very instructive. First, although we have presented the topography as a type of landscape, the participants decided to create symbolic (rather than abstract) representation of what they do. Next, although we prompted participants to challenge or move around flags, we did not observe this happening. Instead of building on each other, which was the original idea, the participants almost “competed” to identify which flag is missing, instead of adding their model in proximity to another flag. As an example, one participant intensively studied the flags and then exclaimed “ah, eco-feminism is missing!”, as if the task was to add what is missing, instead of building on each other. Another interesting aspect was time. As the topography changed over time, many participants came back to see the development, trying to find themselves and evaluating their relation to the newly placed clay and flags.

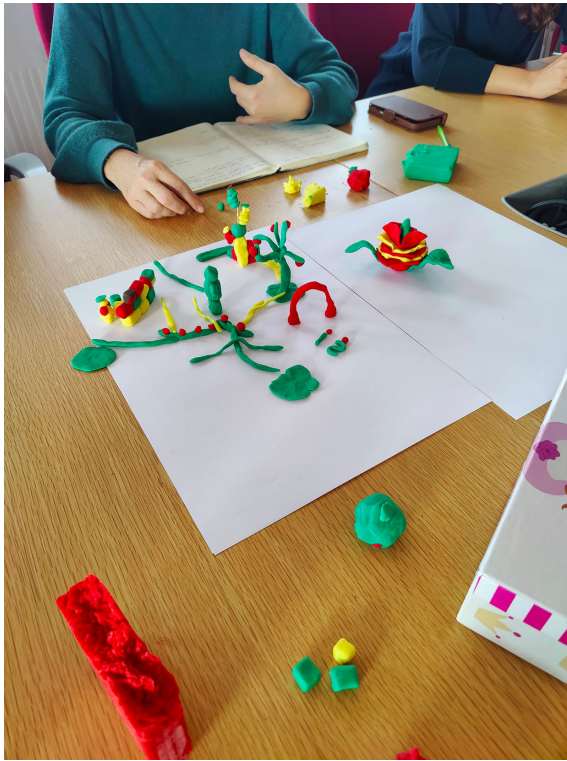
Although the participants did not consider the metaphor of landscape relevant, the physical proximity to the pieces played a role. Participants would position their clay creation close to those they felt were related to them. As the topography grew, someone created a long, tentacle-like “string” made of clay connecting areas that were not physically close. This moment established that it was not only physical proximity that connected the clay pieces.

Most of the pieces made were approached in a relatively non-critical way. Participants would most often wonder about their own positionality within the field of HCI: where they belong and what they are actually doing. There were several exceptions, which pointed to emerging tensions. When looking over the flags, one participant exclaimed “Should we crash Soma design?”. Another participant, when realising a proximity between their clay creation and the keyword “AI”, exclaimed “AI? No!” and quickly lifted their

clay creation and placed it elsewhere. In a similar way, some of the last participants posed several critical questions about the power structures present in our demo: who has the power to decide which clay creations are created? And change these?

Next, we organised a workshop on the Topography of HCI at Halmstad University, as part of a seminar series. This seminar is organised by the research program REBEL, which explores maker pedagogy. Here, we repeated the setup: we provided clay to express one’s position in relation to the research program and maker pedagogy. We introduced the topography and shared insights from the previous topography created at the Sixth Decennial Aarhus Conference, to the six participating researchers (including us as participating organisers). In this workshop, we prompted participants to reflect on their own positionality in relation to the research program and maker pedagogy, and to create a clay-based piece from that reflection. In contrast to the previous iteration, we provided several large chunks of 3-coloured clay that everyone could use as needed. Topography was co-created for about 30 minutes (Figure 2a), after which each participant presented their clay creation. Through these presentations and further reflection, we discussed the created topography and its meanings.

Unsurprisingly, this intimate workshop setting fostered different types of conversations than the large conference floor in Aarhus. We found that the clay served mainly as a starting point for dialogue, helping participants express their positionality regarding the research topic. The relationship between participants also shifted: instead of building sequentially, we created shared meaning and adjusted our sculptures dynamically during the discussion. Interest in “claiming” territory vanished; unlike in Aarhus, no one requested individual flags. We used only one flag to mark the collective piece: “Entrance to REBEL-land” (Figure 2b). Finally, rather than packing up the topography, we left the creations behind as a small exhibition with a sign: “Emergency clay, use it if you need!” Two months



(a) The topography, created at Halmstad University.



(b) Clay pieces on the day of the seminar (28th October 2025).



(c) Clay pieces about two months after the seminar (13th January 2026).

Figure 2: The seminar at Halmstad University.

later, visibly less raw clay remained, suggesting that others using the room had quietly engaged with the materials even after we left (Figure 2c).

2.1 Key Takeaways for CHI 2026

First, we are preparing to manage the tension between constraint and freedom. Our pilot in Aarhus revealed that without clear guidance, participants tended to build isolated symbols rather than adding to a shared terrain, and often prioritised ‘claiming’ territory over connecting to it. Our strategy for CHI is to provide materials that gently structure the interaction, ensuring a cohesive result without excluding individual expression.

Second, we are working with the representation of HCI itself. We are aware that there is no fully accepted definition of HCI, nor is there a list of keywords that capture its ontology. Instead, we are asking participants to choose keywords that represent CHI and help them navigate their relationships with others. We view the categorisation process as an active dialogue, where participants can question or reinforce the boundaries suggested by the clay.

Finally, we are adapting to the massive scale change. The Decennial Aarhus Conference hosted around 300 participants; in comparison, at CHI, it might be around 6000. Moving from hundreds of participants to thousands requires a robust metaphor. We believe the original concept of a ‘topography’ is the most scalable solution;

unlike individual artefacts, which clutter a space, a landscape gains fidelity and richness as more data points (clay) are added. This scale offers a unique opportunity to visualise the ‘weight’ of different subfields in a way previous iterations could not.

3 The CHI 2026 Interactive Experience

The CHI 2026 interactive experience continues our prior work on participatory, material mappings of the HCI field through a shared topography. Building on experiences from the Decennial Aarhus Conference [2] and the workshop at Halmstad University, the topography invites conference participants to collectively shape a physical representation of HCI by placing small amounts of modelling clay on a shared surface. Each contribution reflects a participant’s perspective on the field, including their research focus, methods, theoretical commitments, and areas of interest. Over the course of the conference, these individual contributions accumulate into a collective topography that evolves through ongoing interaction and conversation.

To support the conversation and facilitate overlap among participants’ perspectives, each participant will also have the option to mark a section of the clay topography. Contrary to previous iterations, the marking will be made using a string that, at one end, is connected to the topography and a sticker with a handwritten label on the other (3). We will choose in advance some labels based on

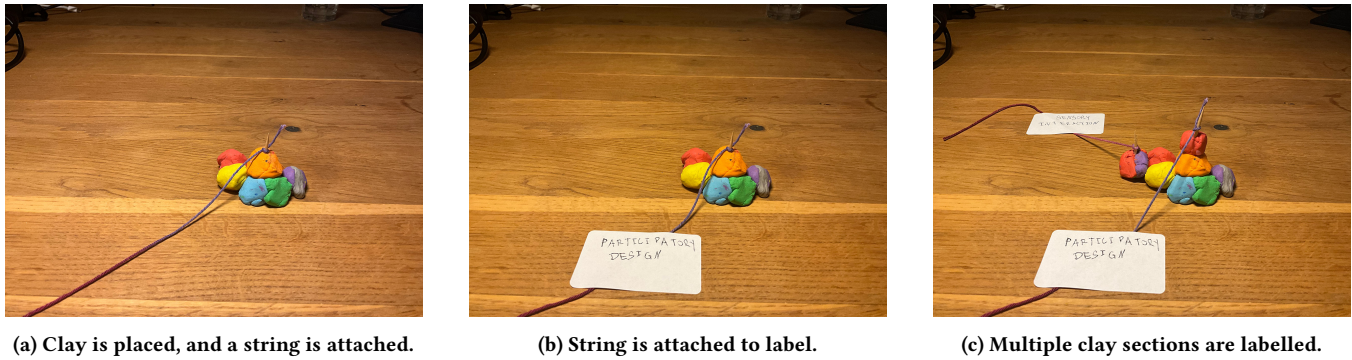


Figure 3: A tiny topography, illustrating the adapted interaction from previous iterations of the demonstrator.

paper keywords from the CHI 2025 proceedings. We hope that this encourages sharing labels, rather than “claiming” one’s keyword or perspective with a flag.

The experience is designed to be lightweight and approachable, allowing participants to engage with it within a short time frame. To accommodate the expected number of participants and mitigate queue formation, the topography supports parallel interaction from multiple sides of the table, enabling several participants to engage simultaneously. This configuration reduces wait time while encouraging simultaneous contributions from participants. The installation is designed to be legible not only to participants but also to bystanders. The evolving topography remains visible throughout the sessions, with annotated flags and a poster providing context for observers. This allows bystanders to grasp the intent of the experience at a glance and to engage informally in conversation, even if they do not immediately contribute. In this way, the experience extends beyond individual interactions, supporting broader, ongoing dialogue about the shape, scope, and shared foundations of CHI.

Beyond its immediate interactivity, the experience is designed to foster reflection on how the CHI community perceives itself. As in earlier iterations, the emerging topography serves as a boundary object that highlights both common ground and differences within the field. Peaks, plateaus, and valleys form through collective action rather than prior categorisation, offering an alternative to established representations of HCI that rely on textual or diagrammatic overviews. The material nature of the topography allows participants to see and feel how their own position relates to others, prompting reflection on proximity, distance, and overlap between subfields. While we provide the material and space for modelling the topography, the participants will remain in control of their engagement; thus, we are excited to see how the topography is shaped in this iteration.

The ongoing interaction and conversation between participants, along with the shared topography, are the main outcomes of our demonstrator. However, we plan to 3D scan the final topography and make it openly available through our website (topography-of-hci.dk), where our past exhibitions of the demonstrator are presented. This allows conference participants and other interested people access to engage with the topography after creation.

4 Conclusion

To conclude, the Topography of HCI offers an alternative way of engaging with our field that is participatory, material, and grounded in collective making. Rather than providing a fixed map of human-computer interaction, this interactive experience invites the CHI community to actively shape a shared representation through clay, conversation, and reflection. Through our iterations in Aarhus and Halmstad, we have learned that the value lies not only in the physical artefact, but equally in the conversations it prompts and the reflections it surfaces. The topography functions as a boundary object that makes visible both the common ground we share and the differences that characterise our diverse community. At CHI 2026, we aim to scale up this experience to accommodate the larger conference community while maintaining its participatory spirit. Whatever shape it ultimately takes, the topography will remain a collective representation of HCI as enacted by those who inhabit it, shaped by the specific moment and context of CHI 2026, and open to continued interpretation and dialogue.

Acknowledgments

Tor-Salve Dalsgaard was supported by the Pioneer Centre for AI [DNRF, grant number P1]. Katerina Cerna was supported by REBEL, a research program at Halmstad University.

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